
ME 601: Advanced Finite Element Methods

Course Code	ME 601
Course Name	Advanced Finite Element Methods
L-T-P-C	3-0-0-3
Intended for	MS/PhD
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Basic concepts**
The standard discrete system, Finite elements of an elastic continuum- displacement approach, Generalization of finite element concepts- weighted residual and variational approaches. [Lectures]
- **Element types**
triangular, rectangular, quadrilateral, sector, curved, isoparametric elements and numerical integration. Automatic mesh generation schemes. [Lectures]
- **Application to structural mechanics problems**
Plane stress and plane strains. Axisymmetric stress analysis, introduction to three dimensional stress analysis. [Lectures]
- **Module IV**
Introduction to use of FEM in steady state field problems- heat conduction fluid flow and non linear material problems, plasticity, creep etc. [Lectures]
- **Module V**
Computer procedure for Finite element analysis. [Lectures]

Text and Reference Books

1. J N Reddy, **An introduction to the Finite Element Method**, McGraw – Hill, New York, 1993.
2. R D Cook, D S Malkus and M E Plesha, **Concepts and Applications of Finite Element Analysis**, 3rd Edition, John Wiley, New York, 1989.
3. K J Bathe, **Finite Element Procedures in Engineering Analysis**, Prentice-Hall, Englewood Cliffs, NJ, 1982.
4. T J R Hughes, **The Finite Element Method**, Prentice-Hall, Englewood Cliffs, NJ, 1986.
5. O C Zienkiewicz and R L Taylor, **The Finite Element Method**, 3rd Edition, McGraw-Hill, 1989.

References

1. Arpaci, V. S., **Conduction Heat Transfer**, Longman Higher Education 1967.
2. Siegel, R., and Howell, J., **Thermal Radiation Heat Transfer**, Taylor and Francis, 2015.
3. Schneider, P., **Conduction Heat Transfer**, Addison-Wesley Pub. Co, 1974.
4. Kakac et al., **Heat Conduction**, 5th Edition, CRC Press, 5th ed, 2018.
5. Myers, **Analytical Methods in Heat Conduction**, 2nd Edition, AMCH, 1998.